BACHELOR OF COMPUTER APPLICATIONS (BCA)

SEMESTER VI - END SEMESTER EXAMINATION

May 2024

Paper Code: BCA-601

Subject: Computer Networks

Time Allowed: 3 Hours | Maximum Marks: 75

General Instructions:

1. Attempt all sections. Section A is compulsory; internal choice is given in Sections B and C.
2. Write answers in legible handwriting and number them exactly as in the question paper.
3. Draw neat, labelled diagrams wherever required.

SECTION A (5 x 3 = 15 Marks)

Answer all five questions. Each question carries 3 marks.

1. Define bandwidth and latency. [3]

2. Differentiate between TCP and UDP. [3]

3. What is a MAC address? [3]

4. Explain the purpose of DNS. [3]

5. List two functions of the network layer. [3]

SECTION B (4 x 10 = 40 Marks)

Answer any four questions. Each question carries 10 marks.

1. Explain the OSI reference model and state the function of each layer. [10]

2. Compare circuit switching and packet switching with suitable examples. [10]

3. Describe CSMA/CD and explain how collisions are handled in Ethernet. [10]

4. Explain IPv4 addressing, subnet masks and subnetting with an example. [10]

SECTION C (2 x 10 = 20 Marks)

Answer any two questions. Each question carries 10 marks.

1. Explain the TCP three-way handshake and connection termination in detail. [10]

2. Discuss the distance vector and link state routing algorithms with neat diagrams. [10]

*** *END OF PAPER* ***